

DIGITAL-TO-ANALOG CONVERSION

DAC

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

DAC Process

Analog Output Demo



DAC



DAC PROCESS

Digital Value

Digital Value = 3102

Decoding

For a 12-bit DAC,

$$\text{DAC} = \frac{\text{Digital Value}}{2^n - 1} \cdot v_{ref}$$

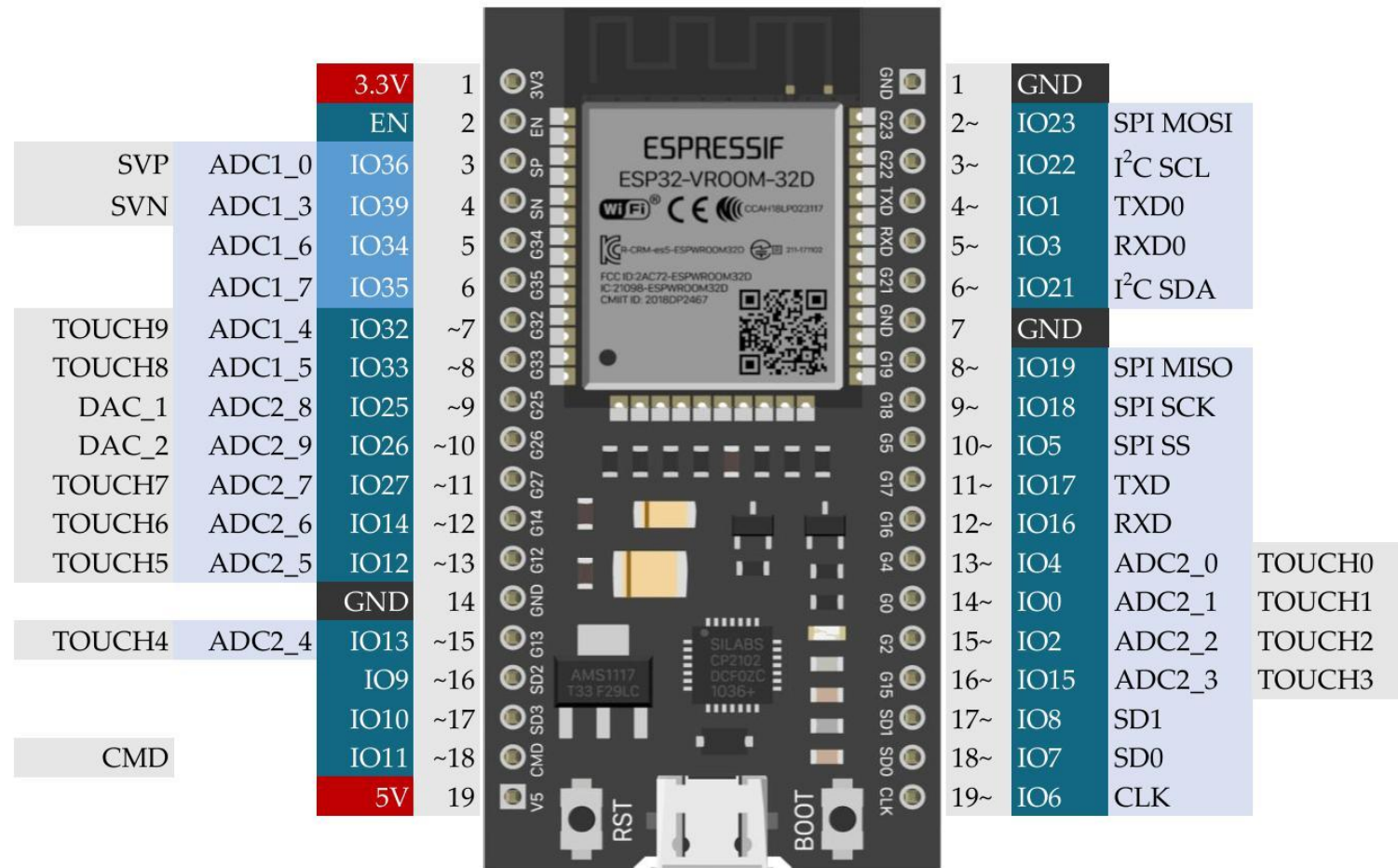
$$\text{DAC} = \frac{3102}{4095} \cdot 3.3$$

Analog Voltage

DAC = 2.5



DEVKITC GPIOs MAP

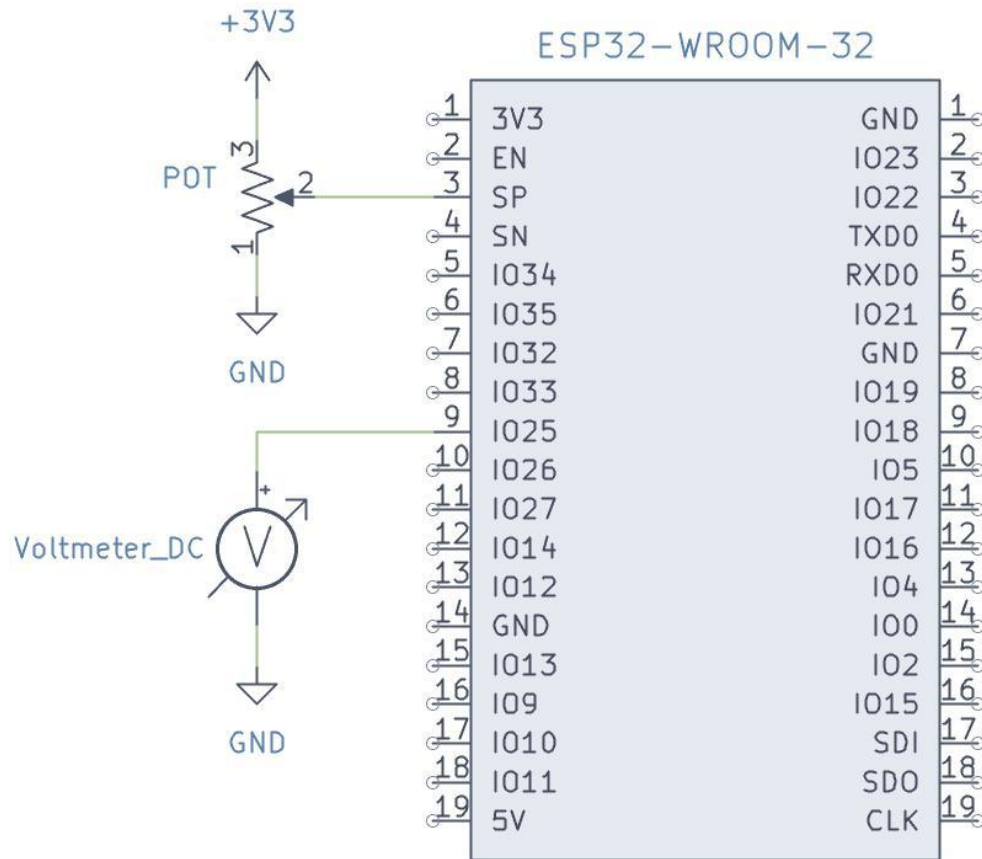


GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

[ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation](#)

ANALOG OUTPUT DEMO



`analogReadResolution()` – configures the precision of the ADC.

ESP32 default ADC resolution is 12-bit.

`dacWrite()` – outputs an analog voltage by converting an **8-bit digital value** into a corresponding voltage level through a DAC pin.



LABORATORY

